

# INTERNSHIP PROJECT REPORT

## REAL-WORLD DATA PROJECT: STRESS-LYSIS HEALTH MONITORING & STRESS LEVEL CLASSIFICATION

|                                                      |                                    |
|------------------------------------------------------|------------------------------------|
| <b>Domain:</b> Healthcare / Health Monitoring        | <b>Organization:</b> Thiranex      |
| <b>Project Type:</b> Machine Learning Classification | <b>Author:</b> Data Science Intern |
| <b>Environment:</b> Python 3.13 / Python IDLE        | <b>Date:</b> October 2026          |

### 1. INTRODUCTION

In recent years, data science and machine learning have become valuable tools across many domains, including health monitoring and physiological data analysis. Modern sensors can continuously record personal measurements such as movement intensity, skin surface conditions, and body temperature. Analyzing these physiological and physical activity indicators allows researchers and engineers to identify hidden patterns and build classification models that categorize stress states.

This project was completed as Task 4 of the Data Science Internship at Thiranex. The project utilizes the Stress-Lysis dataset, which consists of real-world physiological and physical activity readings. The objective is to build an end-to-end machine learning pipeline that explores the data, visualizes feature relationships, scales input variables, trains a Random Forest Classifier, evaluates model accuracy, determines feature importance, and provides an interactive interface for predicting stress levels from new user inputs.

### 2. OBJECTIVES

- **Analyze the Stress-Lysis Dataset:** Load, inspect, and evaluate the statistical structure of the 2,001 data records.
- **Assess Data Integrity:** Verify the shape of the dataset and check for missing or null values.
- **Explore Feature Characteristics:** Compute summary statistics (mean, standard deviation, min, max) for all variables.
- **Visualize Feature Patterns:** Generate exploratory visualizations for target distribution, correlation structures, and boxplots.
- **Build a Machine Learning Classifier:** Train a Random Forest Classifier to assign input records into three stress categories.
- **Evaluate Performance:** Measure performance on an independent test set using accuracy, precision, recall, and F1-score.
- **Analyze Feature Importance:** Quantify how much each feature contributes to the Random Forest decision process.

### 3. DATASET DESCRIPTION

The Stress-Lysis dataset contains 2,001 rows and 4 columns (3 numerical input features and 1 target variable) with 0 missing values across all records.

| Column Name  | Feature Type    | Units              | Description                      |
|--------------|-----------------|--------------------|----------------------------------|
| Humidity     | Numerical Input | mg/min             | Sweat / humidity measurement     |
| Temperature  | Numerical Input | °F                 | Body temperature measurement     |
| Step count   | Numerical Input | steps/min          | Physical activity rate           |
| Stress Level | Target Class    | Category (0, 1, 2) | Classified stress level category |

## 4. MODEL PERFORMANCE & METRICS

The trained Random Forest Classifier achieved a test classification accuracy of 99.83% on 601 test records.

| Stress Category   | Precision | Recall | F1-Score | Support |
|-------------------|-----------|--------|----------|---------|
| Low Stress (0)    | 0.99      | 1.00   | 1.00     | 151     |
| Normal Stress (1) | 1.00      | 1.00   | 1.00     | 237     |
| High Stress (2)   | 1.00      | 1.00   | 1.00     | 213     |
| Overall Accuracy  | —         | —      | 99.83%   | 601     |

## 5. FEATURE IMPORTANCE BREAKDOWN

Random Forest feature importance analysis reveals that Body Temperature and Humidity are the primary drivers of stress classification decisions:

- Temperature: 41.23% importance
- Humidity: 40.15% importance
- Step Count: 18.63% importance

## 6. CONCLUSION

Task 4 of the Data Science Internship at Thiranex successfully demonstrated an end-to-end data science pipeline using the Stress-Lysis dataset. The Random Forest model achieved 99.83% accuracy on unseen test data, driven primarily by body temperature and skin humidity measurements. The system highlights the potential of wearable telemetry analysis in health monitoring applications.